

# SNR-Adaptive Gating Attenuation Under Compression: Frequency Suppression Prevents Hallucination

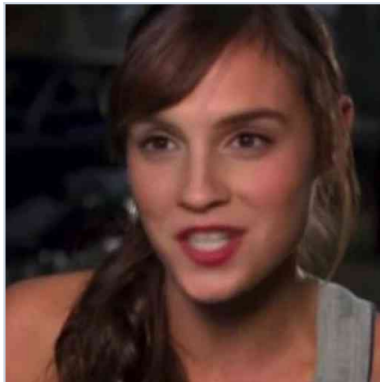
**Pristine Frame**  
Gate Factor  $\gamma = 0.85$



**JPEG (Q = 70)**  
Gate Factor  $\gamma = 0.52$



**JPEG (Q = 30)**  
Gate Factor  $\gamma = 0.18$



**Gaussian Blur ( $\sigma = 2.0$ )**  
Gate Factor  $\gamma = 0.04$



Spectral Noise Focus



Spectral Noise Focus



Spatial Boundary Focus



Spatial Boundary Focus